VISION ENCODER · HALLOWEEN 자율세션 2026.10.30 · NO. 3/4

Perception Encoder: The best visual embeddings are not at the output of the network

One contrastive recipe, every task — if you align with the *right layers*

> **Central claim** — After a scaled image pretraining recipe + robust video data engine, **contrastive vision-language training alone** yields strong, general embeddings for classification, captioning, and localization. The catch: they hide in the **intermediate layers**, so two alignment methods draw them out.

Daniel Bolya · Po-Yao Huang · Peize Sun · et al. (Meta) — `arXiv:2504.13181 (v2)` · Apr 2025

86.6	76.9	94.6 / 80.9 / 82.7	66.0
avg zero-shot ImageNet robustness	zero-shot Kinetics-400 video	DocVQA / InfographicVQA / PerceptionTest · 8B LLM	COCO box mAP · new SotA

Observation — best embeddings = mid-layers › **PE recipe** — contrastive VL + two alignments › **One family** — classification · QA · spatial

01 · PROBLEM One encoder per task is a tax

Traditionally each downstream family wanted its own pretraining objective — classification, captioning, localization each pulled the encoder's features to a different "best" layer, so a single output embedding could not serve all.

PE's reframing: train **one** contrastive vision-language recipe at scale, and the general-purpose embedding exists — just **not where the output head looks**.

02 · METHOD Draw the embedding out

image pretraining recipe · scaled › video data engine › contrastive VL training › 2 alignments → heads

- **Language alignment** — for multimodal language modeling (document/image/video Q&A with an LLM on top).
- **Spatial alignment** — for dense prediction (detection, tracking, depth estimation).

03 · PROBING & LAYERS Deep-dive evidence

- **Table 8** — PEcore frozen features probed with typical probing methods; quality holds across probing protocols.
- **Fig 8** — layer-by-layer analysis: intermediate layers of each pretraining-type (captioning, DINO-style, CLIP-style) evaluated as frozen features — *where* the useful embedding lives depends on the recipe.
- **Figs 9–12** — probing and downstream scalability analyses back the headline results.

04 · READING NOTES Simplification story

A zoo of tailored pretraining objectives collapses into **one contrastive recipe + two alignment heads**. Released: models, code, and PVD — a dataset of **synthetically and human-annotated videos**. The video data engine recaptioned video samples from 0M (image-only baseline) up to **17M samples** during scaling (Fig. 6).

05 · RESULTS Best-in-class across three families

86.6	76.9	66.0
zero-shot ImageNet robustness (avg)	zero-shot Kinetics-400	COCO box mAP — new SotA
94.6	**80.9**	**82.7**
DocVQA	InfographicVQA	PerceptionTest

Q&A numbers come from **8B-scale MLLM** stacks (Llama 3.1-8B / Qwen2.5-7B variants — Tables 10·11); classification numbers are **zero-shot**; the 66.0 COCO box mAP is a **system-level comparison on COCO val2017** (Table 15).

06 · HOW TO READ Three benchmark families, three protocols

86.6 / 76.9 — zero-shot classification & retrieval (image robustness avg on ImageNet variants, video on Kinetics-400) · **94.6 / 80.9 / 82.7** — document/image/video Q&A through an MLLM pipeline with an 8B LLM · **66.0** — system-level COCO detection against leading COCO results (Table 15). Not one leaderboard — three.

07 · FAMILY S/B/L — one recipe, scaled

The family spans model steps **S/14 · B/14 · L/14**; Fig. 2 tunes the pretraining recipe on a fixed data sweep, and Figs. 3–4 show the recipe's gains persist across model sizes (S/B/L) and training steps (120K → 360K).

> **Why contrastive alone is enough here:** the recipe + video engine create the alignment conditions; the two alignment heads then route the same features to language or spatial heads. One backbone, no per-task pretraining detours.

08 · HEAD-TO-HEAD Table 5 · matched 0.1B / 224px scale

Zero-shot	SigLIP2-B	PEcore-B
ObjectNet (IN overlap 113)	73.6	71.9
ObjectNet (all 313)	59.1	58.3
iNaturalist'17	16.9	**25.9**
Dollar St 58	**55.9**	52.1
TextCaps img→txt	72.0	**72.3**
TextCaps flip img→txt	69.8	**71.9**
PVD bench t→v	53.9	**59.8**
PVD bench v→t	60.1	**61.1**

Paper Table 5 rows, transcribed verbatim. At matched tiny scale results are mixed — headline best-in-class claims are at larger family scales.

09 · INSIGHT Where the good embedding hides

The loaded insight for anyone reusing vision encoders: **don't tap the last layer by default**. Fig. 8 evaluates intermediate layers as frozen features across tasks for different pretraining methods — the sweet layer shifts by objective, and PE's contrastive recipe moves it predictably; alignment heads do the routing.

> **Why "language gap" matters:** the language alignment exists precisely because raw mid-layer embeddings are not language-ready; the alignment is the bridge that keeps both dialogue and spatial heads served.

10 · OPEN LIMITS What remains unresolved?

Latency/FLOP trade-offs per family member (Table 4 configurations list sizes; cost curves are in the paper's appendix scales), and the optimal alignment layer per model size is an empirical map (Figs. 8·10–12), not a closed formula. Best-at-scale claims are bounded to the evaluated task set.

11 · SESSION Team poster #3 of 4

할로윈 자율세션 · 팀 공유 논문 포스터 4종 중 세 번째 · Meta.

Reference

Daniel Bolya, Po-Yao Huang, Peize Sun, Jang Hyun Cho, Andrea Madotto, Chen Wei, Tengyu Ma, Jiale Zhi, Jathushan Rajasegaran, Hanoona Rasheed, Junke Wang, Marco Monteiro, Hu Xu, Shiyu Dong, Nikhila Ravi, Daniel Li, Piotr Dollár, Christoph Feichtenhofer. "Perception Encoder: The best visual embeddings are not at the output of the network." arXiv:2504.13181 (v2). 헤드라인 수치는 abstract, 세부 수치는 Table 5에서 전사·재확인 (2026-10-30 자율세션 · ID 검증 원칙).